

// Day 45 of 60

SYSTEM DESIGN

E-Commerce Checkout System

→ Swipe through all 6 slides



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The Hardest Part of E-Comm

Showing products is easy. Money isn't.

Easy:

- Show products
- Add to cart
- Browse categories

Hard:

- Lock inventory atomically
- Handle failed payments
- Coordinate microservices
- Stay correct under load



Checkout is where systems break



Step 1: Inventory Lock

Reserve before you charge

```
User clicks 'Checkout'
```

↓

```
Lock item in DB for 10 min
```

```
available_qty -= 1
```

```
reserved_qty += 1
```

```
reserved_until = now + 10min
```

```
// Without this lock:
```

```
// 2 buyers, 1 last iPhone
```

```
// = angry customer
```



Always reserve before processing payment



Step 2: Payment Intent

Never touch credit cards

Backend → Stripe API:

`'Create intent for $50'`

Stripe → returns:

`client_secret`

Frontend uses secret

to render card form

card data goes to Stripe

(never your servers)

`// PCI compliance: solved.`



Outsource the regulated parts



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Step 3: The Webhook

Don't wait. Get notified.

Bad: backend waits for
Stripe to finish
(timeout risk)

Good: expose webhook URL

Stripe → POST /webhook
payment.succeeded
payment_id: 123

// Async, reliable, retryable



Verify webhook signatures!



Step 4: SAGA Transaction

Coordinate across services

Webhook fires →

Order Service:

confirm order

Inventory Service:

deduct permanently

Notification Service:

send receipt email

// If any step fails:

// compensate (refund/restock)



Distributed transactions need rollback paths



What If Checkout Fails?

Cleanup with cron jobs

User abandons checkout:

Cron job (every 1 min):

```
find reserved items where  
    reserved_until < now  
→ available_qty += 1  
    reserved_qty -= 1
```

// Inventory unlocks

// Someone else can buy it



Idempotency is non-negotiable here



done = true

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